TUTORIAL 5

Linear Systems, Row Reduction, and Invertible Matrices

NOTATION AND CONVENTIONS

For $A \in M_{m\times n}(F)$, the homogeneous solution set is denoted by

$$\ker A = \{x \in F^n : Ax = 0\}.$$

Here F^n is the set of columns of length n over the field F, and 0 on the right is the zero column of length m. The rank $\operatorname{rank} A$ is the number of leading entries in its reduced row echelon form. A variable whose column has no leading entry is a free variable.

MOTIVATION AND GUIDING IDEA

Row reduction replaces a problem with an equivalent one that is easier to read. After calculating, return to the question: is the system consistent, which variables are free, and what is the solution set? For a nonhomogeneous system, the difference of two solutions solves the associated homogeneous system. This explains the form ``a particular solution plus a homogeneous solution.''

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5.1 Linear systems and row reduction

5.1.1 A system of linear equations

RECALL AND TOOLS

In general, for a positive integer n, a linear equation in n unknowns (variables) has the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

Here $a_1, a_2, ..., a_n$ and b are **given scalars** in a specified field $\mathbb{F}$, such as $\mathbb{R}$, $\mathbb{C}$, or a finite field, and $x_1, x_2, ..., x_n$ are the **unknowns**, or variables.
The scalar a_i is the **coefficient** of x_i, and b is the **constant term** of the equation.

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A system of m equations in n unknowns has the general form

$$\begin{cases} a_{1,1}x_1 + a_{1,2}x_2 + \cdots + a_{1,n}x_n = b_1 \\ a_{2,1}x_1 + a_{2,2}x_2 + \cdots + a_{2,n}x_n = b_2 \\ \vdots \\ a_{m,1}x_1 + a_{m,2}x_2 + \cdots + a_{m,n}x_n = b_m \end{cases}$$

To solve the system, it is convenient to collect the **coefficients** in a matrix, preserving the variable order. To shorten the notation, we write either the matrix of variable coefficients, called the **coefficient matrix**, or the **augmented matrix**, which also includes the constant terms. Following the basic rules consistently makes it clear which coefficient belongs to each variable.

EXAMPLE 5.1

$$\begin{cases} 3x_1 - 2x_2 = 0, \\ -6x_1 + 5x_2 = 2 \end{cases} \longleftrightarrow \left(\begin{array}{cc|c} 3 & -2 & 0 \\ -6 & 5 & 2 \end{array}\right).$$

5.1.2 Elementary operations

DEFINITION 5.1

Elementary row operations:
An elementary row operation modifies a matrix in one of the following ways:

1. Interchange rows i and j, denoted by $R_i \longleftrightarrow R_j$.
2. Multiply row i by a **nonzero** scalar λ, denoted by $R_i \longleftarrow \lambda R_i$.
   (Dividing by a nonzero scalar a is multiplication by $1/a$.)
3. Add λ times row j to a different row i, where $i \neq j$, denoted by $R_i \longleftarrow R_i + \lambda R_j$.

- Elementary row operations preserve the solutions of a linear system.
- Every elementary row operation has an inverse that is also an elementary row operation.

DEFINITION 5.2

Matrices A and B are **row equivalent** (or simply equivalent), denoted by $A \sim B$, if B is obtained by applying a finite sequence of elementary row operations to A.

DEFINITION 5.3

Leading entry (pivot): in every nonzero row of a matrix, the first nonzero entry is called its leading entry.

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DEFINITION 5.4

Row echelon form: each leading entry is strictly to the left of the leading entry in the next nonzero row. Any zero rows appear below all nonzero rows.

DEFINITION 5.5

Reduced row echelon form: a matrix in row echelon form such that, in each nonzero row, the leading entry equals 1, and every other entry above and below it in the same column equals 0. These requirements apply to every column, including the constant column of an augmented matrix.

PROPOSITION 5.1

Every matrix is row equivalent to a unique matrix in reduced row echelon form.

Examples:

1. $\begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix}$ is not in echelon form: its leading entries are in the same column.
2. $\begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 3 \end{pmatrix}$ is in echelon form but not reduced echelon form.
3. $\begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 4 \end{pmatrix}$ is in reduced row echelon form.

5.1.3 Gaussian elimination

RECALL AND TOOLS

1. Start with the first active row and the first column. Look in that column for a nonzero entry in the active row or below it. If none exists, move to the next column.
2. Interchange rows to bring this entry into the active row. Division by it is permitted because it has been checked to be nonzero.
3. Multiply the active row by the inverse of its leading entry to obtain a leading 1, and use it to eliminate the entries below.
4. Move to the next active row and seek a pivot column to the right of the previous one. A pivot column need not have the same index as its row.
5. After reaching row echelon form, eliminate the entries above each pivot, working from the last pivot to the first, to obtain reduced row echelon form. This completes Gauss-Jordan elimination.

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EXERCISE 5.1

Please reduce the augmented matrix over $\mathbb{R}$

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 2 & 1 & 3 & 12 \\ 1 & 3 & 2 & 3 \end{array}\right)$$

to reduced row echelon form and please find the system's solution.

WORKED SOLUTION

Apply $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - R_1$:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & -3 & 1 & 10 \\ 0 & 1 & 1 & 2 \end{array}\right).$$

Swap the last two rows and apply $R_3 \leftarrow R_3 + 3R_2$:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 4 & 16 \end{array}\right).$$

Divide R_3 by 4, subtract it from R_2, and clear the entries above the last two pivots:

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 4 \end{array}\right).$$

The unique solution is $(1, -2, 4)^T$. Substitution gives $1-4+4 = 1$, $2-2+12 = 12$, $1-6+8 = 3$.

NOTE

To solve a linear system, reduce its **augmented matrix**, including the constant terms. You may stop at row echelon form and use back substitution, or continue to reduced row echelon form and read off the solutions.

EXAMPLE 5.2

Consider the system

$$\begin{cases} x + 2y = 0, \\ y + z = 2, \\ 2x + z = 7. \end{cases}$$

Its augmented matrix is

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 1 & 1 & 2 \\ 2 & 0 & 1 & 7 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 5 & 15 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{array}\right).$$

The first step uses $R_3 \leftarrow R_3 - 2R_1 + 4R_2$. Hence $z = 3$, $y = 2 - z = -1$ and $x = -2y = 2$.

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The solution set is $\{(2,-1,3)\}$.

This example shows how row reduction helps solve a linear system. We will next see that a system need not have a unique solution.

EXERCISE 5.2

Please reduce the augmented matrix over $\mathbb{Z}_{11}$

$$\left(\begin{array}{cccc|c} 1 & 0 & 2 & 1 & 3 \\ 0 & 1 & 3 & 2 & 4 \\ 2 & 3 & 4 & 1 & 5 \end{array}\right)$$

to reduced row echelon form. Please perform the arithmetic modulo 11.

WORKED SOLUTION

Apply $R_3 \leftarrow R_3 - 2R_1 - 3R_2$. The last row has entries $4-4-9 = -9 \equiv 2$, $1-2-6 = -7 \equiv 4$ and $5-6-12 = -13 \equiv 9$. Thus

$$\left(\begin{array}{cccc|c} 1 & 0 & 2 & 1 & 3 \\ 0 & 1 & 3 & 2 & 4 \\ 0 & 0 & 2 & 4 & 9 \end{array}\right).$$

The inverse of 2 is 6. Multiplying R_3 by 6 gives

$$\left(\begin{array}{cccc|c} 1 & 0 & 2 & 1 & 3 \\ 0 & 1 & 3 & 2 & 4 \\ 0 & 0 & 1 & 2 & 10 \end{array}\right).$$

Finally apply $R_1 \leftarrow R_1 - 2R_3$ and $R_2 \leftarrow R_2 - 3R_3$:

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & 8 & 5 \\ 0 & 1 & 0 & 7 & 7 \\ 0 & 0 & 1 & 2 & 10 \end{array}\right).$$

All entries use representatives $0, 1, \ldots, 10$.

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EXERCISE 5.3

Please solve the following system over $\mathbb{C}$:

$$\begin{cases} iz + (2-i)w = -3-i, \\ 2iz + (5-2i)w = -5-5i. \end{cases}$$

Please verify the solution in both equations.

WORKED SOLUTION

Subtract twice the first equation from the second to get

$$w = (-5-5i) - 2(-3-i) = 1-3i.$$

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EXERCISE 5.3 CONTINUED

Now $(2-i)(1-3i) = -1-7i$, so

$$iz = -3 - i - (-1-7i) = -2+6i, \qquad z = 6+2i.$$

The first equation gives $(-2+6i)+(-1-7i) = -3-i$. The second gives

$$2iz + (5-2i)w = (-4+12i)+(-1-17i) = -5-5i.$$

5.2 Solving linear systems

DEFINITION 5.6

After row reduction, a variable whose column in the coefficient matrix contains a leading entry is called a **pivot variable** (or dependent variable). A variable whose column contains no leading entry is called a **free variable**. In a consistent system, every free variable can be assigned an arbitrary value in the field, after which the pivot variables are determined by the equations. Over an infinite field, a free parameter gives infinitely many choices; over a finite field, the number of choices is finite. The number of free variables is the number of degrees of freedom of a consistent system.

DEFINITION 5.7

The rank of a matrix A, denoted by $\operatorname{rank}(A)$, is the number of nonzero rows in a row echelon form. Equivalently, it is the number of leading entries (pivots) in that form.

Remark: if a system with coefficient matrix A and n variables is consistent, its number of degrees of freedom is $n - \operatorname{rank}(A)$. Over $\mathbb{F}_q$, it has exactly $q^{n-\operatorname{rank}(A)}$ solutions.

EXERCISE 5.4

Please solve over $\mathbb{R}$ the system

$$\begin{cases} x + 3y - 2z = 7, \\ 2x - y + 3z = 0, \\ 4x + 5y - z = 14. \end{cases}$$

WORKED SOLUTION

Write the augmented matrix and subtract multiples of its first row:

$$\left(\begin{array}{ccc|c} 1 & 3 & -2 & 7 \\ 2 & -1 & 3 & 0 \\ 4 & 5 & -1 & 14 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 3 & -2 & 7 \\ 0 & -7 & 7 & -14 \\ 0 & -7 & 7 & -14 \end{array}\right).$$

Subtract row two from row three, divide row two by -7 and clear its pivot column:

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right).$$

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EXERCISE 5.4 CONTINUED

Put $z = t$. Then $x = 1 - t$, $y = 2 + t$, so

$$S = (1, 2, 0) + \operatorname{span}\{(-1, 1, 1)\}.$$

There is one free variable and infinitely many solutions. Substitution gives $7, 0, 14$ in the three equations for every $t \in \mathbb{R}$.

EXERCISE 5.5

Please solve the system

$$\begin{cases} 2x_1 - 6x_2 + 4x_3 = 7, \\ x_1 - 3x_2 + 2x_3 = 2. \end{cases}$$

WORKED SOLUTION

Swap the rows, then subtract twice row one from row two:

$$\left(\begin{array}{ccc|c} 2 & -6 & 4 & 7 \\ 1 & -3 & 2 & 2 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & -3 & 2 & 2 \\ 0 & 0 & 0 & 3 \end{array}\right).$$

The last row says $0 = 3$, so there is no solution.

Geometric illustration

We have encountered three possible cases:

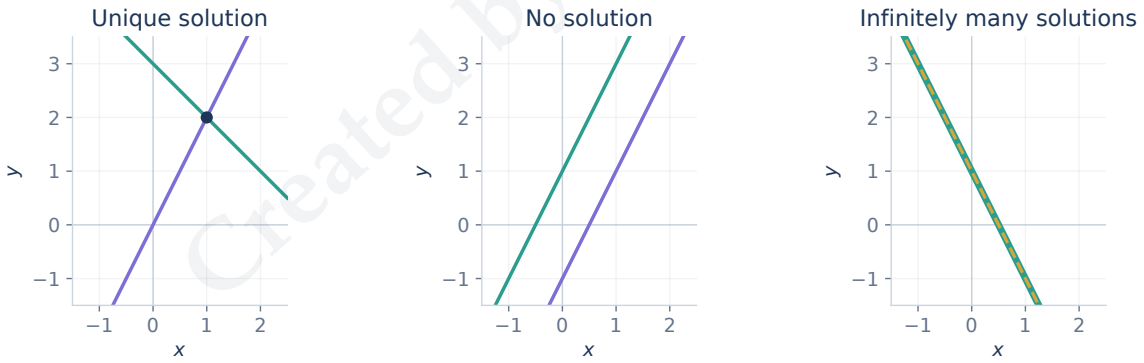


Figure 5.1. Over the real numbers, two equations in two unknowns can describe intersecting lines, distinct parallel lines, or coincident lines. These give, respectively, a unique solution, no solution, or infinitely many solutions.

EXERCISE 5.6

Let $a \in \mathbb{R}$. Please classify the number of solutions and please state the coefficient matrix rank in each case:

$$\begin{cases} x - 2y = 0, \\ 3x + 3z = 2, \\ x + az = 1. \end{cases}$$

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EXERCISE 5.6 CONTINUED

WORKED SOLUTION

Subtract three times row one from row two, and row one from row three. Swap the last two rows and subtract three times row two from row three:

$$\left(\begin{array}{ccc|c} 1 & -2 & 0 & 0 \\ 3 & 0 & 3 & 2 \\ 1 & 0 & a & 1 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & -2 & 0 & 0 \\ 0 & 2 & a & 1 \\ 0 & 0 & 3-3a & -1 \end{array}\right).$$

If $a = 1$, there is a contradiction row and rank 2. If $a \neq 1$, the rank is 3 and the solution is unique:

$$z = -\frac{1}{3-3a}, \qquad x = 1 + \frac{a}{3-3a}, \qquad y = \frac{1}{2}\left(1 + \frac{a}{3-3a}\right).$$

No value of a gives infinitely many solutions.

5.2.1 Homogeneous systems

RECALL AND TOOLS

All constant terms are 0. In this case:

- We may write only the coefficient matrix rather than the augmented matrix: the zero column is unchanged by all row operations, so there is no need to keep writing it.
- A solution always exists: no contradiction row can occur, so inconsistency is impossible. We must still determine whether the zero solution is the only solution. Over an infinite field, the existence of a nonzero solution implies infinitely many solutions.
- If there is a unique solution, it is the zero solution, also called the **trivial solution**.
- The solution set of a homogeneous system in n variables is a vector subspace of $\mathbb{F}^n$ and can always be expressed as a span.

EXERCISE 5.7

Let $a \in \mathbb{R}$. Please classify the number of solutions of this homogeneous system and please find the solution set in each case:

$$\begin{cases} x - 2y = 0, \\ 3x + 3z = 0, \\ x + az = 0. \end{cases}$$

WORKED SOLUTION

The coefficient matrix is the same as in the preceding exercise. Row reduction gives

$$\begin{pmatrix} 1 & -2 & 0 \\ 3 & 0 & 3 \\ 1 & 0 & a \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & 0 \\ 0 & 2 & a \\ 0 & 0 & 3-3a \end{pmatrix}.$$

If $a \neq 1$, the rank is 3 and only the zero solution exists. If $a = 1$, the rank is 2, and $x = -z$,

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EXERCISE 5.7 CONTINUED

$y = -z/2$. Choosing $z = 2t$ gives

$$S = \{(-2t, -t, 2t) : t \in \mathbb{R}\} = \text{span}\{(-2, -1, 2)\}.$$

There are infinitely many solutions in this case. A homogeneous system is always consistent, so no case has zero solutions.

5.3 Supplementary exercises: measurements and counting solutions

EXERCISE 5.8

Consider the system

$$x_1 + x_2 = 7, \qquad x_2 + x_3 = 11, \qquad x_1 + x_3 = 8.$$

Please find x_1, x_2, x_3. Please find the only value of c for which adding $x_1 + x_2 + x_3 = c$ leaves the solution set unchanged. Please determine what happens for $c = 14$.

WORKED SOLUTION

Add the first two equations and subtract the third:

$$2x_2 = 7 + 11 - 8 = 10, \qquad x_2 = 5.$$

Then $x_1 = 7 - 5 = 2$ and $x_3 = 11 - 5 = 6$. The third equation checks as $2 + 6 = 8$. Adding all three equations gives

$$2(x_1 + x_2 + x_3) = 7 + 11 + 8 = 26.$$

Thus the required value is $c = 13$. Setting $c = 14$ creates a contradiction, so the augmented system has no solution.

EXERCISE 5.9

Please solve over $\mathbb{Z}_3$ the system

$$\begin{cases} x + y + z = 1, \\ x + z = 2. \end{cases}$$

Please count its solutions and please compare with the same equations over $\mathbb{R}$.

WORKED SOLUTION

Subtracting the second equation from the first gives $y = -1 = 2$ in $\mathbb{Z}_3$. Choosing $z = t$ gives

$$(x, y, z) = (2 - t, 2, t), \qquad t \in \mathbb{Z}_3.$$

The three solutions are $(2, 2, 0), (1, 2, 1), (0, 2, 2)$. Over $\mathbb{R}$ we get

$$(x, y, z) = (2 - t, -1, t), \qquad t \in \mathbb{R},$$

so there are infinitely many solutions. Both systems have one free variable, but the number of solutions depends on the field size.

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Additional practice

EXERCISE 5.10

Let $p(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3$ be a real polynomial.

1. Please find all polynomials satisfying $p(0) = 0$, $p(2) = 4$, $p'(0) = 0$, $p'(1) = 0$. Please determine whether the solution set is a subspace and whether $t = 1$ is a local maximum or minimum.
2. Please repeat with $p(0) = 2$, $p(2) = 6$, $p'(0) = 0$, $p'(1) = 0$.
3. Add $p'(2) = 5$ to the second part. Please determine whether the system is consistent and whether its solution set is a subspace.

WORKED SOLUTION

We have $p'(t) = a_1 + 2a_2 t + 3a_3 t^2$. In part one, $a_0 = a_1 = 0$, while the remaining conditions give

$$4a_2 + 8a_3 = 4, \qquad 2a_2 + 3a_3 = 0.$$

Subtracting twice the second equation from the first gives $2a_3 = 4$, so $a_3 = 2$ and $a_2 = -3$. The unique solution is

$$p_1(t) = -3t^2 + 2t^3.$$

Check: $p_1(2) = -12 + 16 = 4$ and $p_1'(1) = -6 + 6 = 0$. The zero polynomial fails $p(2) = 4$, so the solution set is not a subspace. Also $p_1''(t) = -6 + 12t$, hence $p_1''(1) = 6 > 0$, giving a local minimum.
In part two, $a_0 = 2$, $a_1 = 0$, and the other equations are unchanged. Thus

$$p_2(t) = 2 - 3t^2 + 2t^3.$$

Again the solution is unique, its set is not a subspace, and $t = 1$ is a local minimum. In part three, this unique polynomial gives

$$p_2'(2) = -12 + 24 = 12 \neq 5.$$

The system is inconsistent, and the empty solution set is not a subspace.

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EXERCISE 5.11

Let

$$p(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 \in \mathbb{R}_3[t].$$

1. Please find all polynomials satisfying $p(0) = 0$, $p(1) = -1$, and $p'(0) = p'(1/2) = 0$. Please determine whether the solution set is a subspace and whether $t = 1/2$ is a local maximum or minimum.
2. Please repeat for $p(0) = 2$, $p(1) = 1$, and $p'(0) = p'(1/2) = 0$.
3. Please add $p'(1) = 1$ to the preceding conditions and please determine whether a solution exists.

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EXERCISE 5.11 CONTINUED

WORKED SOLUTION

$$p'(t) = a_1 + 2a_2t + 3a_3t^2.$$

In part 1, $a_0 = a_1 = 0$. The remaining conditions give

$$a_2 + a_3 = -1, \qquad a_2 + \tfrac{3}{4}a_3 = 0.$$

Subtracting gives $a_3/4 = -1$, so $a_3 = -4$ and $a_2 = 3$. The unique solution is

$$p_1(t) = 3t^2 - 4t^3.$$

Indeed, $p_1(1) = -1$ and $p_1'(1/2) = 3 - 3 = 0$. The solution set is not a subspace because the zero polynomial fails $p(1) = -1$. Also,

$$p_1''(t) = 6 - 24t, \qquad p_1''(1/2) = -6 < 0,$$

so this is a local maximum.
In part 2, $a_0 = 2$, $a_1 = 0$, and $a_2 + a_3 = 1 - 2 = -1$. The same system determines a_2, a_3, giving

$$p_2(t) = 2 + 3t^2 - 4t^3.$$

The solution is again unique, and its set is not a subspace because it does not contain the zero polynomial. The derivatives coincide, so $t = 1/2$ is again a local maximum.
In part 3, $p_2'(1) = 6 - 12 = -6 \neq 1$. Thus the additional condition contradicts the preceding ones. The solution set is empty and is not a subspace.

5.4 Invertible Matrices

DEFINITION 5.8

A matrix $A \in M_n(\mathbb{F})$, with $n \geq 1$, is **invertible** if there is $B \in M_n(\mathbb{F})$ satisfying $AB = BA = I_n$.

The unique inverse B is denoted A^{-1}.

Basic Properties

RECALL AND TOOLS

- If A is invertible, so is A^{-1}.
- If A, B are invertible, so is AB, and $(AB)^{-1} = B^{-1}A^{-1}$.
- If A is invertible, so is A^T, and $(A^T)^{-1} = (A^{-1})^T$.
- The identity is invertible, with $I^{-1} = I$.
- A square zero matrix of positive order is not invertible.

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5.4.1 Exercises on the Algebra of Invertible Matrices

EXERCISE 5.12

Let A, B be square matrices, with A invertible, such that $A = B^2$ and $B = A^2$. Please prove that B is the inverse of A, that is, $B = A^{-1}$.

WORKED SOLUTION

We have

$$\begin{aligned} AB &= AA^2 = A^3, \\ A^3 &= A^3(AA^{-1}) = A^4A^{-1} \\ &= (A^2A^2)A^{-1} = B^2A^{-1} = AA^{-1} = I. \end{aligned}$$

Since $B = A^2$, also $BA = A^2A = A^3 = AB = I$. Therefore $AB = BA = I$, and $B = A^{-1}$.

EXERCISE 5.13

Let A, B be square matrices such that AB is invertible. Please prove that both A and B are invertible. **This is the converse of the product assertion from the lecture.**

WORKED SOLUTION

Use the fact that for square matrices of the same order, a one-sided inverse is also a two-sided inverse. Indeed, $AX = I$ implies $n = \text{rank}(AX) \leq \text{rank}(A) \leq n$, so A is invertible. From $(AB)(AB)^{-1} = I$, obtain $A[B(AB)^{-1}] = I$. Thus A is invertible, with inverse $B(AB)^{-1}$. Similarly, $(AB)^{-1}(AB) = I$ gives $[(AB)^{-1}A]B = I$. Thus B is invertible, with inverse $(AB)^{-1}A$.

5.4.2 An Algorithm for Finding the Inverse

PROPOSITION 5.2

A square matrix A is invertible if and only if it is row-equivalent to the identity matrix.

Algorithm

1. Form $(A \mid I)$ and row-reduce, aiming to put the left block into **reduced row-echelon form**.
2. Inspect the resulting left block:
   - If it is I, then A is invertible, and the right block is its inverse: $(I \mid A^{-1})$.
   - If it is not I, then A is not invertible.

EXERCISE 5.14

Please determine which matrices are invertible. For each invertible matrix, please find its inverse by row reduction and please verify the result.

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & 6 \\ -2 & -4 \end{pmatrix}.$$

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EXERCISE 5.14 CONTINUED

WORKED SOLUTION

$$\left(\begin{array}{ccc|ccc} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & -2 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right).$$

$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 + 2R_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -2 & 1 & 2 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right).$$

$$A^{-1} = \begin{pmatrix} -2 & 1 & 2 \\ 1 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix}, \qquad AA^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -2+2 & 1 & 2-2 \\ 1-1 & 0 & 1 \end{pmatrix} = I_3.$$

In B, the second row is $-2/3$ times the first. The operation $R_2 \leftarrow R_2 + (2/3)R_1$ creates a zero row, so B is not invertible.

5.5 Elementary Matrices

DEFINITION 5.9

- Elementary matrices are square matrices representing the elementary row operations used in row reduction.
- Applying an elementary row operation to a matrix is equivalent to multiplying it **on the left** by the corresponding elementary matrix.
- Each elementary matrix is obtained by applying its row operation to the identity matrix.

Consequence. Matrices A, B are row-equivalent if and only if there is a finite sequence of elementary matrices such that $B = E_1 E_2 \cdots E_k A$.

1. **Interchanging rows** i and j is represented by $E^{(1)}_{i,j}$:

$$E^{(1)}_{i,j} = \begin{pmatrix} 1 & & & & \\ & 0 & & 1 & \\ & & 1 & & \\ & 1 & & 0 & \\ & & & & 1 \end{pmatrix}.$$

   - This matrix is its own inverse: interchange the same two rows again.

2. **Multiplying row** i by a nonzero scalar λ is represented by $E^{(2)}_{i,\lambda}$:

$$E^{(2)}_{i,\lambda} = \begin{pmatrix} 1 & & & & \\ & \ddots & & & \\ & & \lambda & & \\ & & & \ddots & \\ & & & & 1 \end{pmatrix}.$$

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- Its inverse is $E^{(2)}_{i,\lambda^{-1}}$: multiply the same row by the reciprocal scalar.

3. **Adding λ times row j** to row i, with $i \neq j$, is represented by $E^{(3)}_{i,j,\lambda}$. The entry λ is in row i, column j:

$$E^{(3)}_{i,j,\lambda} = \begin{pmatrix} 1 & & & & & \\ & \ddots & & & & \\ & & 1 & & \lambda & \\ & & & \ddots & & \\ & & & & 1 & \\ & & & & & 1 \end{pmatrix}.$$

- Its inverse is $E^{(3)}_{i,j,-\lambda}$: add $-\lambda$ times row j to row i.

EXERCISE 5.15

Please use the preceding reduction to write $A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix}$ as a product of elementary matrices.

WORKED SOLUTION

The reduction operations are $R_1 \leftrightarrow R_2$, then $R_3 \leftarrow R_3 - R_2$, then $R_1 \leftarrow R_1 + 2R_3$. Their matrices are

$$E_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad E_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}, \quad E_3 = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

$$E_3E_2E_1A = I_3, \qquad A = E_1^{-1}E_2^{-1}E_3^{-1}$$
$$= \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The last two factors multiply to $\begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$. The first factor swaps the first two rows and recovers A. Each factor is elementary.

EXERCISE 5.16

Let $A, B, C \in \mathbb{R}^{n\times n}$ be invertible, and let $\alpha \in \mathbb{R}$ with $\alpha \neq 0$. Define

$$D := \alpha A^3 B^T C^2 A.$$

1. Please prove that D is invertible.
2. Please find an explicit expression for D^{-1} in terms of A, B, C, α.

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EXERCISE 5.16 CONTINUED

WORKED SOLUTION

1. Each of $\alpha I_n, A, A^3, B^T, C^2$ is invertible:

   - $\alpha \neq 0$, so αI_n is invertible.
   - A is invertible by assumption, so A^3 is invertible.
   - Since B is invertible, B^T is invertible with $(B^T)^{-1} = (B^{-1})^T$.
   - Since C is invertible, C^2 is invertible with $(C^2)^{-1} = (C^{-1})^2$.

   A product of invertible matrices is invertible, so D is invertible.

2. Write the ordered product $D = \alpha A^3 B^T C^2 A$. The inverse of a product is the product of the inverses in reverse order:

$$(XYZ)^{-1} = Z^{-1}Y^{-1}X^{-1}.$$

   Thus

$$D^{-1} = A^{-1}(C^2)^{-1}(B^T)^{-1}(A^3)^{-1}(\alpha I_n)^{-1}.$$

   Compute each factor:

$$(C^2)^{-1} = (C^{-1})^2, \quad (B^T)^{-1} = (B^{-1})^T, \quad (A^3)^{-1} = (A^{-1})^3, \quad (\alpha I_n)^{-1} = \frac{1}{\alpha} I_n.$$

   Substitution gives

$$D^{-1} = A^{-1}(C^{-1})^2(B^{-1})^T(A^{-1})^3 \cdot \frac{1}{\alpha} I_n = \frac{1}{\alpha} A^{-1}(C^{-1})^2(B^{-1})^T(A^{-1})^3.$$

   This is the required expression for D^{-1}.

EXERCISE 5.17

Let

$$A = \operatorname{diag}(\lambda_1, \ldots, \lambda_n) = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} \in \mathbb{R}^{n \times n}.$$

Please prove that A is invertible if and only if every $\lambda_i \neq 0$, and please find an explicit formula for A^{-1}.

WORKED SOLUTION

If A is invertible, then every $\lambda_i \neq 0$. Suppose A is invertible and, for a contradiction, that

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EXERCISE 5.17 CONTINUED

$\lambda_k = 0$ for some index k. Consider the standard vector

$$\vec{e}_k = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix},$$

with 1 in position k and zero in every other coordinate. Then

$$A\vec{e}_k = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ \lambda_k \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \vec{0},$$

since $\lambda_k = 0$. Thus $A\vec{e}_k = \vec{0}$ with $\vec{e}_k \neq \vec{0}$, a nontrivial solution of $Ax = 0$. If A were invertible, left multiplication by A^{-1} would give $e_k = A^{-1}0 = 0$, a contradiction. Therefore every $\lambda_i \neq 0$. **If every $\lambda_i \neq 0$, then A is invertible; we find its inverse.** Assume all these scalars are nonzero. Define

$$B := \operatorname{diag}\left(\frac{1}{\lambda_1}, \ldots, \frac{1}{\lambda_n}\right) = \begin{pmatrix} \frac{1}{\lambda_1} & 0 & \cdots & 0 \\ 0 & \frac{1}{\lambda_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{\lambda_n} \end{pmatrix}.$$

Calculate

$$\begin{aligned} AB &= \operatorname{diag}(\lambda_1, \ldots, \lambda_n) \operatorname{diag}\left(\frac{1}{\lambda_1}, \ldots, \frac{1}{\lambda_n}\right) \\ &= \operatorname{diag}\left(\lambda_1 \frac{1}{\lambda_1}, \ldots, \lambda_n \frac{1}{\lambda_n}\right) \\ &= \operatorname{diag}(1, \ldots, 1) = I_n. \end{aligned}$$

The same calculation in reverse order gives $BA = I_n$. Hence B is the inverse of A, so

$$A^{-1} = B = \operatorname{diag}\left(\frac{1}{\lambda_1}, \ldots, \frac{1}{\lambda_n}\right).$$

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5.5.1 Equivalent Conditions

PROPOSITION 5.3

For a square matrix $A \in M_n(\mathbb{F})$, the following conditions are equivalent:

1. A is invertible.
2. A is row-equivalent to I_n.
3. $\text{rank}(A) = n$.
4. For every column vector $\vec{b} \in \mathbb{F}^n$, the system $A\vec{x} = \vec{b}$ has a solution.
5. For every column vector $\vec{b} \in \mathbb{F}^n$, the system $A\vec{x} = \vec{b}$ has a unique solution $\vec{x} \in \mathbb{F}^n$.

5.5.2 The Connection with the Homogeneous System

In the preceding assertions, take $\vec{b} = \vec{0}$. There is then a unique solution, the trivial solution. Thus:

PROPOSITION 5.4

Another equivalent condition: $A \in M_n(\mathbb{R})$ is invertible if and only if its homogeneous system has only the trivial solution. In other words, the only solution of $A\vec{x} = \vec{0}$ is $\vec{x} = \vec{0}$.

EXERCISE 5.18

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix}, \qquad b = (\alpha, 2\gamma, 3\phi)^T, \qquad \alpha, \gamma, \phi \in \mathbb{R}.$$

Please solve $Ax = b$ in terms of the parameters and please prove that the system is homogeneous if and only if its unique solution is zero.

WORKED SOLUTION

$$x = A^{-1}b = \begin{pmatrix} -2 & 1 & 2 \\ 1 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} \alpha \\ 2\gamma \\ 3\phi \end{pmatrix} = \begin{pmatrix} -2\alpha + 2\gamma + 6\phi \\ \alpha \\ -\alpha + 3\phi \end{pmatrix}.$$

Indeed, $x_2 = \alpha$, $x_1 - 2x_3 = 2\gamma$, and $x_2 + x_3 = 3\phi$. The solution is unique because A is invertible. The system is homogeneous exactly when $b = 0$, equivalently $\alpha = \gamma = \phi = 0$, which gives $x = 0$. Conversely, if $x = 0$ is a solution, then $b = A0 = 0$.

EXERCISE 5.19

$$A = I_3 + N, \qquad N = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{pmatrix}.$$

Please compute N^2, N^3, please find A^{-1} without row reduction, and please solve $Ax = (3, 8, 3)^T$.

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EXERCISE 5.19 CONTINUED

WORKED SOLUTION

$$N^2 = \begin{pmatrix} 0 & 0 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad N^3 = 0.$$

$$(I+N)(I-N+N^2) = I - N + N^2 + N - N^2 + N^3 = I,$$

$$(I-N+N^2)(I+N) = I + N - N - N^2 + N^2 + N^3 = I.$$

$$A^{-1} = I - N + N^2 = \begin{pmatrix} 1 & 1 & -3 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{pmatrix}.$$

$$x = A^{-1}\begin{pmatrix} 3 \\ 8 \\ 3 \end{pmatrix} = \begin{pmatrix} 3+8-3\cdot 3 \\ 8-3\cdot 3 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}.$$

The only nonzero entry of N^2 is $(-1)\cdot 3 = -3$. Verification gives $Ax = (2-(-1), -1+3\cdot 3, 3)^T = (3,8,3)^T$.

Additional practice

EXERCISE 5.20 CHALLENGE

Let $u, v \in \mathbb{R}^n$, $n \geq 1$, and define $A = I_n + uv^T$.

1. Please prove that A is invertible if and only if $1 + v^T u \neq 0$, and please find its inverse in this case without using a determinant.
2. Please apply the formula to $u = (1,-1,0)^T$, $v = (2,0,1)^T$, and please solve $Ax = (5,0,-1)^T$.

WORKED SOLUTION

Write $P = uv^T$ and $c = v^T u \in \mathbb{R}$. Associativity gives

$$P^2 = (uv^T)(uv^T) = u(v^T u)v^T = cP.$$

If $1 + c \neq 0$, define $B = I_n - P/(1+c)$. Then

$$AB = (I_n + P)\left(I_n - \frac{P}{1+c}\right) = I_n + \left(1 - \frac{1}{1+c} - \frac{c}{1+c}\right)P = I_n.$$

Multiplication in the opposite order gives exactly the same expansion, so $BA = I_n$. Therefore

$$A^{-1} = I_n - \frac{uv^T}{1+v^T u}.$$

If $1 + c = 0$, then $c = -1$, so $u \neq 0$, whereas

$$Au = u + u(v^T u) = (1+c)u = 0.$$

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EXERCISE 5.20 CHALLENGE CONTINUED

An inverse would imply $u = 0$, a contradiction. This proves both directions.
For the given example,

$$v^T u = 2, \quad A = \begin{pmatrix} 3 & 0 & 1 \\ -2 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}, \qquad A^{-1} = \begin{pmatrix} \frac{1}{3} & 0 & -\frac{1}{3} \\ \frac{2}{3} & 1 & \frac{1}{3} \\ 0 & 0 & 1 \end{pmatrix}.$$

$$x = A^{-1} \begin{pmatrix} 5 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} (5+1)/3 \\ 10/3 - 1/3 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}.$$

Verification: $Ax = (3 \cdot 2 - 1, -2 \cdot 2 + 3 + 1, -1)^T = (5, 0, -1)^T$.

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EXERCISE 5.21 CHALLENGE

Let $A \in M_n(\mathbb{R})$ satisfy $A^m = I_n$ for an integer $m \geq 2$, and define

$$P = \frac{1}{m} \sum_{k=0}^{m-1} A^k.$$

Please prove that $AP = PA = P$ and $P^2 = P$. Please find the inverse of $I_n + P$.

WORKED SOLUTION

Multiplication by A shifts the powers, so

$$AP = \frac{1}{m}(A + A^2 + \cdots + A^m) = P.$$

Since powers of A commute with A, we also have $PA = P$. Induction gives $A^k P = P$ for every $k \geq 0$. Thus

$$P^2 = \frac{1}{m} \sum_{k=0}^{m-1} A^k P = P.$$

Consequently

$$(I_n + P)(I_n - P/2) = I_n + P/2 - P^2/2 = I_n.$$

The reverse product is also I_n, since both factors are polynomials in P. Therefore $(I_n+P)^{-1} = I_n - P/2$.

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EXERCISE 5.22 CHALLENGE

Let $A, B \in M_n(\mathbb{R})$ and define the block matrix

$$C = \begin{pmatrix} A & B \\ B & A \end{pmatrix}.$$

Please prove that C is invertible if and only if $A+B$ and $A-B$ are invertible. In that case, please find C^{-1} in terms of $(A+B)^{-1}$ and $(A-B)^{-1}$.

WORKED SOLUTION

Define

$$J = \begin{pmatrix} I_n & I_n \\ I_n & -I_n \end{pmatrix}.$$

Since $J^2 = 2I_{2n}$, we have $J^{-1} = J/2$. Block multiplication gives

$$CJ = \begin{pmatrix} A+B & A-B \\ A+B & -(A-B) \end{pmatrix},$$

and therefore

$$J^{-1}CJ = \begin{pmatrix} A+B & 0 \\ 0 & A-B \end{pmatrix}.$$

Multiplication on either side by invertible matrices preserves invertibility. A block diagonal matrix is invertible exactly when both its diagonal blocks are invertible. This proves the equivalence. Writing $X = (A+B)^{-1}$ and $Y = (A-B)^{-1}$, we obtain

$$\begin{aligned} C^{-1} &= J \begin{pmatrix} X & 0 \\ 0 & Y \end{pmatrix} J^{-1} \\ &= \frac{1}{2} \begin{pmatrix} X+Y & X-Y \\ X-Y & X+Y \end{pmatrix}. \end{aligned}$$

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EXERCISE 5.23 CHALLENGE

Consider a finite linear system over a field F with $n \geq 1$ unknowns. Assume every collection of at most $n+1$ of its equations is consistent. Please prove that the full system is consistent. Please construct an inconsistent system of $n+1$ equations for which every collection of n equations is consistent.

WORKED SOLUTION

Suppose the system is inconsistent and choose an inconsistent collection of minimum size k. Write its equations as

$$\sum_{i=1}^{n} a_{ji} x_i = b_j, \qquad 1 \leq j \leq k.$$

The hypothesis gives $k > n+1$. Consider the homogeneous system in $\lambda_1, \ldots, \lambda_k$ given by

$$\sum_{j=1}^{k} \lambda_j a_{ji} = 0 \quad (1 \leq i \leq n), \qquad \sum_{j=1}^{k} \lambda_j b_j = 0.$$

It has $n+1$ equations and more than $n+1$ unknowns. Row reduction therefore gives a nonzero solution. Choose j with $\lambda_j \neq 0$. These identities express equation j, including its right-hand side, as a linear combination of the other equations. Hence every solution of the other equations also satisfies equation j. Minimality of k says that the other equations have a solution. That solution satisfies the whole collection, a contradiction.
The bound is sharp. Take

$$x_1 = 0, \ldots, x_n = 0, \qquad x_1 + \cdots + x_n = 1.$$

The full system is inconsistent. If the final equation is omitted, zero is a solution. If $x_j = 0$ is omitted, setting $x_j = 1$ and all other unknowns to zero gives a solution. Thus every collection of n equations is consistent.

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